# Gullak: learn money by living it, with play money

An adaptive, multilingual financial-literacy PWA for Indian students in Classes 6–12. Live at **gullak.abgupta.me**; open source (MIT).

> **Guardrail, enforced in code:** Gullak educates and never recommends a real product or tells anyone to buy or sell. Every projection lists its assumptions. All money is virtual: the portfolio type is `isVirtual: true` as a literal, so real money cannot be represented, and the database enforces it too.

## 1. Problem and our solution

Most Indian school students first meet money decisions (UPI, scams, saving, inflation, investing) with no safe place to practise them. Textbook finance is abstract, English-first and built for one-off reading, not for habits. Students often share a family phone with poor connectivity.

Gullak teaches through a single loop: **experience → decide → see the consequence → understand → apply → review → unlock**. Learning sits in **eight connected worlds**: Candy Economy, Savings Quest, Bankopolis, Market Mayhem, Grow Your Money, Stock Street, Life Simulator and Finance Tycoon. The worlds run from pocket money to banking, markets, investing, life finance and entrepreneurship. All grade groups share one map; the class group (6–7, 8–9, 10–12) changes depth, vocabulary and realism, not the course.

What sets it apart is **reach and habit, not games alone**:

- **Practice on real history.** Episodes replay real Indian market data with life events, then reveal what really happened.
- **Take-home missions** make the learner the family's money teacher.
- **Built for shared phones and no data:** learner profiles on the device and offline lessons.
- **The real market, fully mapped for a teenager:** all **61 SEBI mutual-fund categories**, each with a plain-language summary and a "watch out" note, plus **111 stock-market metrics** explained on fictional companies. Few school products go beyond "what is a share".
- **A habit loop that is never addictive:** no loot boxes, guilt notifications or returns leaderboards.

### How a learner uses it

- **Learn by deciding, not reading.** Each 3–5 minute lesson opens with a familiar situation, such as pocket money, a shop or a scam message, and asks the learner to choose. They see the consequence first, and only then is the idea named and explained. A quick application and a quiz follow, and passing the quiz unlocks the next topic on the world map.
- **Hold a portfolio of play money.** Every learner starts with virtual cash scaled to their class group. Their **paper-trading portfolio** (gullak.abgupta.me/portfolio) holds play-money cash and holdings, and Practice decisions flow into it. The backend already values portfolios on real dated prices and NAVs, opens virtual fixed deposits and serves market quotes with their source. Every projection shows its assumptions, so it reads as a lesson, never a promise.
- **Live through a simulation.** A Practice episode replays real history month by month, together with life events. In the flagship episode the market falls in March 2020 just as a hospital bill arrives. Learners without an emergency fund must sell at that day's low price. A "rewind" then shows the same months with savings set aside, and the reveal says "FYI, this really happened." Liquidity and emergency funds are learned by feeling them.
- **Take it home and come back.** Take-home quests ask the learner to teach a family member what they learned, so one phone reaches the whole household. XP, levels and a gentle day streak reward the habit. Gullu, the piggy-bank companion, gives tips that fit where the learner is and explains any money term on tap.
- **Adapts to the learner.** Quiz answers and decisions update a per-concept mastery estimate. The next recommendation either revisits the weakest idea or opens the topic the learner is most ready for, and it always says why.

## 2. Approach and architecture

| Layer | What it does |
|---|---|
| Web app (Next.js 16 PWA) | One responsive app for phone and desktop: a bottom tab bar and touch targets on phones, a sidebar and multi-pane layout on desktop. Installable on Android, iOS, Windows and macOS. A service worker (Serwist) caches lessons for offline use. English, Hindi and Telugu through `next-intl`, with the locale in a cookie for a stable start URL. |
| Engine (pure TypeScript) | Compound interest, FD, RD and SIP maths in integer paise. Portfolio valuation, goal planning, starting capital, unlock rules, unit-price comparison, the lesson-format picker, a recommendation ranker and the knowledge-tracing runtime. No framework imports, so it is fully unit-tested. |
| Content as data | A 33-topic prerequisite graph mapped onto the eight worlds. Each topic has lessons and quizzes, plus spot-and-sort rounds, branching stories and Practice episodes. Everything is Zod-validated at load: no cycles, no unknown prerequisites, identical keys in every language. A new lesson, world or language needs no renderer code. |

| | |
|---|---|
| Real data | Adapters for AMFI NAVs, NSE index history and bank FD rates, each with a cache and a committed snapshot fallback. The demo never depends on a live API, and every value shows its source and as-of date. |
| Backend (Supabase) | A verified-session API for onboarding, lesson progress, server-graded quizzes, virtual FDs, portfolio valuation, activity and episode runs with resumable bookmarks, and mission completion. Content hashes are pinned and writes are idempotent. Row-level security protects every learner table, and all access goes through one typed data-access layer. |
| AI and adaptivity | A provider-agnostic LLM client (Gemini, falling back to Groq, falling back to content search) with versioned prompts and guardrails that screen learner questions and model output for advice in all three languages. Knowledge tracing is trained in Colab and exported as JSON for the TypeScript engine. Lesson generation is offline: drafts are validated against curriculum briefs, cached, and reviewed by a human before they are used. |

**Adaptivity.** Onboarding asks only for a starting point: the learner's experience sets a prior mastery of 0.1, 0.3 or 0.5. After that, evidence from quizzes, activities and decisions drives mastery. Two pure engine functions use it:

- `recommendNext` first reviews the weakest concept in a passed topic. Otherwise it picks the open topic whose prerequisites are best mastered, and it explains its choice with a reason code.
- `pickLessonVariant` follows the learner's preferred formats in order, or rotates them with "Mix it up". It never changes unlocks.

Passing a quiz is the only unlock rule, so the system can always explain it to a learner, a parent or a teacher.

## 3. Features built

### Learning

- **Onboarding** in en / hi / te. The learner picks a language, class group, experience and formats. Starting play money scales with the class group (₹2,000 to ₹20,000).
- **World map** of the eight worlds. It scrolls to the current world, gives one clear "Play" action and explains locked or coming-soon worlds accessibly.
- **Lessons and quizzes.** An 8-step lesson renderer. Passing a quiz unlocks the next topics, filtered by class group.
- **Mechanics:** spot-and-sort rounds, branching story choices, a money-term decoder, a virtual shop (Candy Economy) and a life-simulator mission.
- **Recommended next:** a "because…" card driven by the mastery-based ranker.

### Practice, money and habit

- **Practice episode:** the March 2020 crash plus a hospital bill on real NIFTY 50 index-fund history. Selling is forced at the day's price, and a "rewind" comparison shows what an emergency fund would have changed.
- **Paper trading** at `/portfolio`: play-money cash and holdings, backed by a tested portfolio API that values holdings on real dated prices and opens virtual FDs. There is also a wallet-and-savings-jars game with a two-business risk comparison. The play-money badge is on every money screen.
- **Take-home quests** to do with family, and real progress: XP from passed topics and activities, levels, and a day streak that is hidden at zero.
- **Gullu the companion:** contextual tips in three languages, "What's this?" explanations for terms, and a `/help` page.
- **Safety:** confirmation before destructive actions, restored quiz drafts, save retries, error and not-found pages, and separate profiles for each learner on a shared device.

### Real-world depth

| What | Coverage |
|---|---|
| Stock-market terms | **111 company metrics** explained for learners, the full set a stock-research site shows. They fall into 10 groups: price and size, results, growth, balance sheet, cash flow, returns and efficiency, debt and safety, valuation and ownership, plus how to read time windows (quarter, TTM, 3/5/10-year averages and CAGR). Each entry says what the metric tells you, how it is calculated and which class group sees it. Written as the content source for Stock Street's company report card (next to build), taught on fictional companies only, with no buy or sell cut-offs. |
| Market and fund types | **61 SEBI mutual-fund categories**, each taught with a plain-language summary ("lends money for just one day at a time") and a "watch out" note on its risk. They fall into 8 groups: debt (18), equity (11), sector and thematic (11), hybrid (8), index funds and ETFs (7), fund of funds (3), solution-oriented (2) and specialised (1). Each one is tagged with its instrument kind, risk band and the topic that teaches it. |
| Curriculum | **33 topics** in a prerequisite graph across the eight worlds. They run from needs versus wants and budgeting, through UPI, phishing, fake loan apps and reporting fraud, to inflation, compounding, FDs, RDs, PPF, gold, bonds, debt funds, SIPs, index funds and market ups and downs. Each topic has a lesson and a quiz, and is tagged with the class groups it suits. |
| Real Indian data | Daily AMFI NAV history for a NIFTY 50 index fund since 2019, which includes the March 2020 crash, the NIFTY 50 index series and a dated bank one-year FD rate. Each comes with its source and as-of date, behind adapters with caching and a snapshot fallback. |

## 4. Results

### Knowledge tracing: real-data benchmark

We compared candidate mastery models on **283,105 real answers** from 4,163 learners across 149 skills in the public ASSISTments 2009–10 skill-builder dataset (Feng, Heffernan & Koedinger, 2009). We used 5-fold cross-validation, split by learner, with one-step-ahead prediction. The dataset is downloaded at run time and never committed.

| Model | AUC | RMSE | Accuracy | Parameters / runtime |
|---|---|---|---|---|
| Overall correct rate (baseline) | 0.500 | 0.474 | 0.660 | 1 |
| Per-skill correct rate | 0.629 | 0.462 | 0.671 | 1 per skill |
| PFA | 0.710 | 0.444 | 0.703 | Logistic, per skill |
| BKT | 0.722 | 0.438 | 0.715 | 4 per skill |
| **BKT + forgetting (chosen)** | **0.727** | **0.436** | **0.717** | 5 readable per skill; runs offline on a phone |
| DKT (LSTM) | 0.739 | 0.432 | 0.724 | Neural network; needs large data and a GPU |

**Decision:** BKT with forgetting reaches about 98% of DKT's AUC. It has five parameters per concept that a teacher can read, and it runs in plain TypeScript on cheap phones without data. DKT stays a later option once Gullak has learners at scale. Parameters fitted on synthetic learners are labelled as drafts and are never used as the production mastery model.

### Other experiments

- **Format picker:** exhaustive tests over every class group and preference set. Explicit preferences are honoured in order, "Mix it up" rotates formats, quizzes are never chosen as lessons, and missing formats fall back to the authored order with an explicit reason.
- **Real-data replay:** the committed index-fund series reproduces the March 2020 fall and recovery the flagship episode needs. The AMFI parser handles both published column layouts.
- **Backend correctness:** local PostgreSQL tests show that a retried request with the same idempotency key applies once, a stale revision is rejected, and two learners on one shared device never see each other's progress. A hosted acceptance run against the deployed Supabase project is recorded in `docs/`.

### Quality and testing

| Check | Result |
|---|---|
| Continuous integration | On every push and PR, including teammates' forks: commit rules, Biome lint, strict TypeScript, about 560 Vitest unit tests across all packages, a production build, and Playwright end-to-end tests on phone and desktop viewports. A red build is never merged. 21 tagged releases of the v0.2 line so far. |
| Finance engine | Normal, edge and invalid inputs tested for every formula; integer paise throughout, so there is no floating-point drift. |
| Content and languages | Every topic, brief, activity, episode and snapshot is schema-validated in CI, and every locale must have exactly the same message keys. |
| Safety | Guardrails screen tutor input and output for advice in en / hi / te. A UI test checks that no screen calls a product "recommended". The type system cannot represent real money. |
| Design review | A heuristic UX critique of the learner app led to fixes for jargon, focus order, contrast, hidden actions, made-up progress numbers and missing error recovery before release. |

## 5. Limitations and next steps

- **Not yet wired:** progress is stored on the device. The backend API is built and tested, but the learner UI does not sync to it yet. The AI tutor client and its guardrails exist but are not shown in the app. Next: connect the paper-trading buy and sell screens to the portfolio and FD APIs, and build Live mode on current prices and the Stock Street report card.
- **Content migration:** the app still runs topics through the earlier topic model, mapped onto the eight worlds. Full world-and-lesson content, the *Nani's Gullak* storyline arc for each world, and Live paper trading come next.
- **Languages:** Hindi and Telugu interface strings are drafts awaiting human review. Topic content is English-first, with Sarvam translation and review planned.
- **Adaptivity:** activate BKT with forgetting on real Gullak evidence, then show "Why this?" on every recommendation. Build a facilitator dashboard for teachers and NGOs.